Result 1 (homology identity fixes classes). [Mathlib, CategoryTheory.Functor.map_id]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X be a topological space. Let x be an element of $H(X)$. Then

$$\mathrm{id}_{X*}(x) = x.$$

Adapted from the cited source.

Result 2 (homology respects composition). [Mathlib, CategoryTheory.Functor.map_comp]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X, Y and Z be topological spaces. Let f be a continuous map from X to Y. Let g be a continuous map from Y to Z. Let x be an element of $H(X)$. Then

$$(g \circ f)_*(x) = g_*(f_*(x)).$$

Adapted from the cited source.

Result 3 (induced function acts on classes). [Topology foundations, induced homology map definition]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X and Y be topological spaces. Let f be a continuous map from X to Y. Let x be an element of $H(X)$. Let $h = f_*$. Then

$$h(x) = f_*(x).$$

Result 4 (homology image membership). [Mathlib, LinearMap.mem_range]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X and Y be topological spaces. Let f be a continuous map from X to Y. Let x be an element of $H(Y)$. Let $h = f_*$. Let $U = \mathrm{range}(f_*)$. Then

$$x \text{ belongs to } U \text{ if and only if } (\text{there exists } y : H(X),\ h(y) = x)\,.$$

Adapted from the cited source.

Result 5 (bundled class has its declared space). [Topology foundations, computed dependent record field]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let P be a space equipped with a homology class. Then

$$(\text{there exists } x : H(P.\text{space}),\ x = P.\text{element})\,.$$

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.